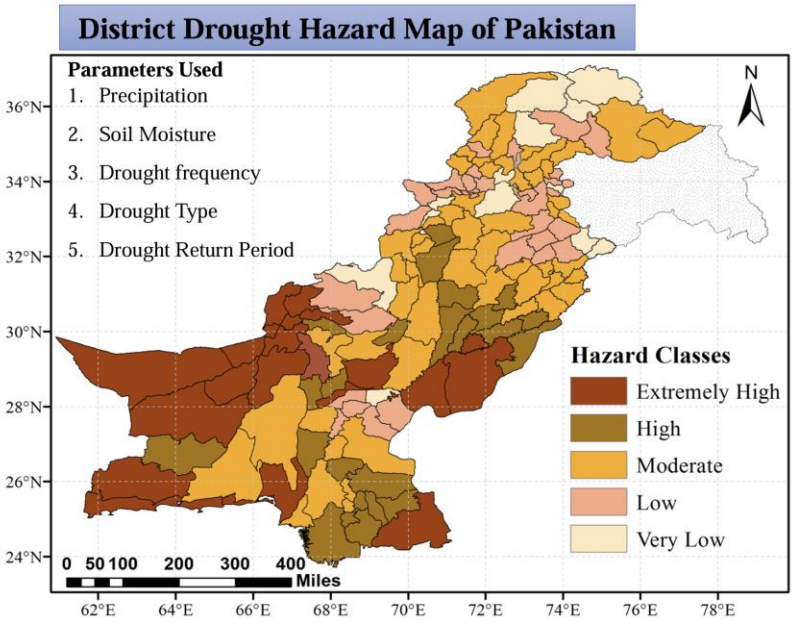


Extremely high vulnerable districts to drought
Total= 19 districts

Sindh

1. Tharparkar

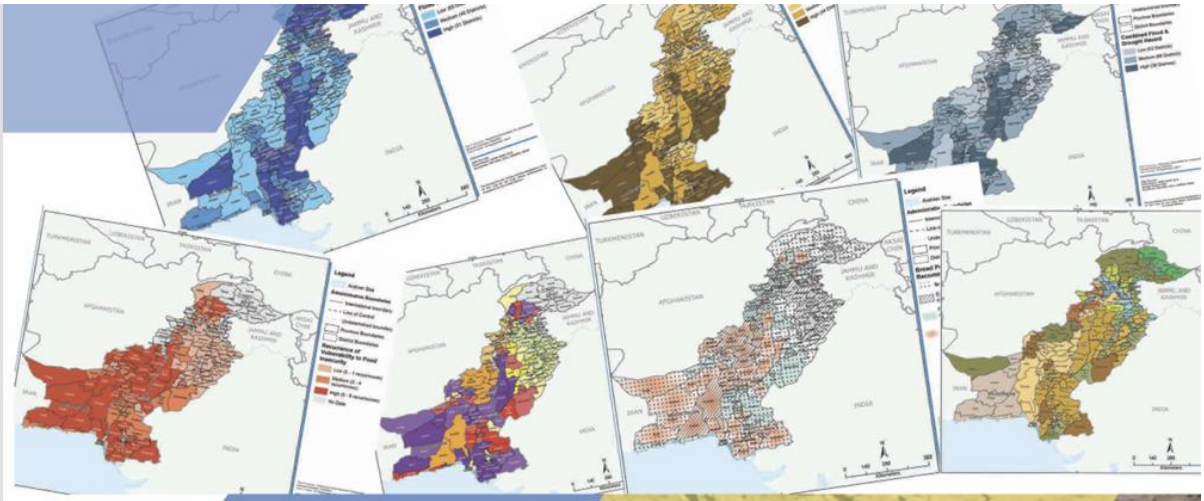
Balochistan

1. Chagi
2. Dera Bugti
3. Gawadar
4. Jaffarabad
5. Kech
6. Kharan
7. Lesbella
8. Mastung
9. Nushki
10. Pishin
11. Qalat
12. Qilla Abdullah
13. Quetta
14. Washuk
15. Ziarat)

Punjab

1. Bahawalpur
2. Lodharn
3. RYKhan

Adnan, S., & Ullah, K. (2020). Development of drought hazard index for vulnerability assessment in Pakistan. *Natural Hazards*, *103*(3), 2989-3010.



INTEGRATED CONTEXT ANALYSIS (ICA)

on Vulnerability to Food Insecurity and Natural Hazards

Pakistan 2017

PAKISTAN National Disaster Management Authority NDMA

WFP wfp.org

Number of Articles Published

S.No.	Paper Name	Status of Journals
1	Latif, M., Zoon, M., **Adnan, S.**, Ahmed, R., Hannachi, A., Mahmood, R., & Umar, M. (2024). Spatiotemporal analyses of temperature and equivalent temperature and their relationship with crop health across Pakistan's cropland. *Theoretical and Applied Climatology*, 1-19.**(IF: 3.409).**	**Q2**
2	Tanveer, S. K., Hussain, I., Abbass, H., Sohail, M., Qamar, M., Mehmood, Z., ... & **Adnan, S.** (2023). WHEAT CROP PRODUCTIVITY IN THE MUNGBEAN-WHEAT CROPPING SYSTEM IN RAINFED AREA OF POTOHAR PAKISTAN. *SAARC Journal of Agriculture*, *21*(2).	
3	Tariq, A., Beni, L. H., Ali, S., **Adnan, S.,**& Hatamleh, W. A. (2023). An effective geospatial-based flash flood susceptibility assessment with hydrogeomorphic responses on groundwater recharge. *Groundwater for Sustainable Development*, 100998.**(IF: 5.9).**	**Q1**
4	Mohsin, M., &**Adnan, S.**(2023) Probabilistic modelling of interarrival time of drought for different operational drought indices used in Pakistan. *International Journal of Climatology*. **(IF: 4.069).**	**Q1**
5	Ahmed, N., Lu, H., Booij, M. J., Wang, G., Marhaento, H., Bhat, M. S., &**Adnan, S.** (2022). Innovative polygon trend analysis of monthly precipitation (1952–2015) in the Hindukush-Karakoram-Himalaya river basins of Pakistan. *International Journal of Climatology*.**(IF: 4.069).**	**Q1**
6	**Adnan, S.,**& Ullah, K. (2022). Long-term trends in climate parameters and multiple indices for drought monitoring over Pakistan. *Meteorology and Atmospheric Physics*, *134*(4), 1-22.**(IF: 2.292).**	**Q3**
7	Iqbal, S. W., Latif, M., Ahmed, R., **Adnan, S.,** Umar, M., Ahmad, B. ...& Syed, J. H. (2022). Performance evaluation and comparison of observed and reanalysis gridded precipitation datasets over Pakistan. *Theoretical and Applied Climatology*, 1-24.**(IF: 3.409).**	**Q2**
8	Ashraf, M., Ullah, K., &**Adnan, S.** (2022). Satellite based impact assessment of temperature and rainfall variability on drought indices in Southern Pakistan. *International Journal of Applied Earth Observation and Geoinformation*, *108*, 102726. **(IF: 7.672).**	**Q1**

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9	Syed, F.S., **Adnan, S.**, Zamreeq, A., Ghulam A. (2022). Identification of droughts over Saudi Arabia and global teleconnections. Nat Hazards. https://doi.org/10.1007/s11069-022-05285-z. **(IF: 3.158).**	**Q1**
10	Durrani, I. H., Adnan, S.,& Aftab, S. M. (2021). Daily Climate Extremes of Temperature and Precipitation Over Quetta Valley, Pakistan During 1961-2019. *International Journal of Environment*, *10*(1), 20-48.	
11	Saddique, A., **Adnan, S.,** Bokhari, H., Azam, A., Rana, M. S., Khan, M. M. ...& Sharif, S. (2021). Prevalence and Associated Risk Factor of COVID-19 and Impacts of Meteorological and Social Variables on Its Propagation in Punjab, Pakistan. *Earth Systems and Environment*, **1-14. (IF: 7.2).**	**Q1**
12	Qaiser, G., Tariq, S., **Adnan, S.**, & Latif, M. (2021).Evaluation of a composite drought index to identify seasonal drought and its associated atmospheric dynamics in Northern Punjab, Pakistan. *Journal of Arid Environments*, *185*, 104332. **(IF: 2.759).**	**Q1**
13	**Adnan, S.,** Hanif, M., Khan, A. H., Latif, M., Ullah, K., Bashir, F., ... & Haider, S. (2020). Impact of Heat Index and Ultraviolet Index on COVID-19 in Major Cities of Pakistan. *Journal of occupational and environmental medicine*.**(IF: 2.162).**	**Q2**
14	Hashmi, M. Z., Ashraf, K., **Adnan, S.**, Wang, S., Pongpiachan, S., Su, X., & Ahmed, Z. (2020). Spatio-temporal assessment and climatology of atmospheric organic carbon over Pakistan. *Arabian Journal of Geosciences*, *13*(15), 1-13.**(IF: 1.827).**	**Q2**
15	**Adnan, S.**, & Ullah, K. (2020). Development of drought hazard index for vulnerability assessment in Pakistan. *Natural Hazards*, 1-22.**(IF: 3.102).**	**Q1**
16	Ali, S. M., Khalid, B., Akhter, A., Islam, A., &**Adnan, S.** (2020). Analyzing the occurrence of floods and droughts in connection with climate change in Punjab province, Pakistan. *Natural Hazards*, *103*(2), 2533-2559.**(IF: 3.102).**	**Q1**
17	Ain, N.U., Latif, M., Ullah, K., **Adnan, S.**, Ahmed, R., Umar, M. and Azam, M., (2020. Investigation of seasonal droughts and related large-scale atmospheric dynamics over the Potwar Plateau of Pakistan. *Theoretical and Applied Climatology*, pp.1-21.**(IF: 3.179).**	**Q2**
18	Ahmed, F., **Adnan, S.,** & Latif, M. (2020). Impact of jet stream and associated mechanisms on winter precipitation in Pakistan. *Meteorology and Atmospheric*	**Q3**

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	Physics, *132*(2), 225-238. **(IF: 2.065)**	
19	Baig, M. H. A., Abid, M., Khan, M. R., Jiao, W., Amin, M., &**Adnan, S.** (2020). Assessing meteorological and agricultural drought in Chitral Kabul river basin using multiple drought indices. *Remote Sensing*, *12*(9), 1417. **(IF: 4.848).**	**Q1**
20	**Adnan, S**., Ullah, K. and Ahmed, R., 2019. Variability in meteorological parameters and their impact on evapotranspiration in a humid zone of Pakistan. *Meteorological Applications*.**(IF: 1.685)**	**Q2**
21	Ahmed, R., Latif, M., **Adnan, S**. and Abuzar, M.K., (2019). Thunderstorm frequency distribution and associated convective mechanisms over Pakistan. *Theoretical and Applied Climatology*, *137*(1-2), pp.755-773.**(IF: 2.882).**	**Q2**
22	Durrani, I. H., **Adnan, S**., & Aftab, S. M. (2018). Historical and Future Climatological Drought Projections Over Quetta Valley, Balochistan, Pakistan. In *IOP Conference Series: Materials Science and Engineering* 414(1): 012043. IOP Publishing. **(IF: 7.328)**	**Q1**
23	**Adnan, S.,** Ullah, K., Shuanglin, L., Gao, S., Khan, A. H., & Mahmood, R. (2018). Comparison of various drought indices to monitor drought status in Pakistan. *Climate Dynamics*, *51*(5), 1885-1899.**(IF: 4.048)**	**Q1**
24	Durrani, I. H., **Adnan, S.,** Ahmad, M., Khair, S. M., & Kakar, E. (2018). Observed long-term climatic variability and its impacts on the ground water level of Quetta alluvial. *Iranian Journal of Science and Technology, Transactions A: Science*, *42*(2), 589-600.**(IF: 0.692)**	**Q2**
25	Gul, S., Hussain, I., Shad, M. Y., Faisal, M., Shoukry, A. M., &**Adnan, S**. (2018). Nonparametric trend analysis of reference evapotranspiration for Khyber Pakhtunkhwa, Pakistan. *International Journal of Global Warming*, *14*(3), 313-329.**(IF: 0.779)**	**Q3**
26	**Adnan, S.,** Ullah, K., Khan, A. H., & Gao, S. (2017). Meteorological impacts on evapotranspiration in different climatic zones of Pakistan. *Journal of Arid Land*, *9*(6), 938-952.**(IF: 1.444)**	**Q2**
27	**Adnan, S.,** Ullah, K., Gao, S., Khosa, A. H., & Wang, Z. (2017). Shifting of agro-climatic zones, their drought vulnerability, and precipitation and temperature trends in Pakistan. *International Journal of Climatology*, *37*, 529-543.**(IF: 3.10)**	**Q1**

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28	**Adnan, S.,** Ullah, K., & Shouting, G. (2016). Investigations into precipitation and drought climatologies in South Central Asia with special focus on Pakistan over the period 1951–2010. *Journal of Climate*, *29*(16), 6019-6035.**(IF: 4.161)**	**Q1**
29	**Adnan, S.,** Ullah, K., & Gao, S. (2015). Characterization of drought and its assessment over Sindh, Pakistan during 1951–2010. *Journal of Meteorological Research*, *29*(5), 837-857.**(IF: 1.116)**	**Q1**
30	Haider, S., & **Adnan, S.**(2014). Classification and assessment of aridity over Pakistan provinces (1960-2009). *International Journal of Environment*, *3*(4), 24-35	
31	Hanif M, Khan AH, &**Adnan S**. (2013). Latitudinal precipitation characteristics and trends in Pakistan. *Journal of Hydrology* 492: 266-272. **(IF: 2.693)**	**Q1**
32	**Adnan, S**., Hayat Khan, A., Haider, S., & Mahmood, R. (2012). Solar energy potential in Pakistan. *Journal of Renewable and Sustainable Energy*, *4*(3), 032701. **(IF: 1.514)**	**Q2**
33	Jaafar, M. Z., Khan, A. H., **Adnan, S.**, Markwitz, A., Siddique, N., Waheed, S., & Brereton, R. G. (2011). Multiblock analysis of environmental measurements: A case study of using Proton Induced X-ray Emission and meteorology dataset obtained from Islamabad Pakistan. *Chemometrics and Intelligent Laboratory Systems*, *107*(1), 31-43.**(IF: 1.92)**	**Q2**
34	**Adnan, S.**, Mahmood, R., & Khan, A. H. (2009). Water Balance Conditions in Rainfed Areas of Potohar and Balochistan Plateau during 1931-08. *World Applied Sciences Journal*, *7*(2), 162-169.	
35	**Adnan, S.**, & Khan, A. H. (2009). Effective rainfall for irrigated agriculture plains of Pakistan. *Pak J Meteorol*, *6*(11), 61-72.	

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Shahzada Adnan

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Solar energy potential in Pakistan S Adnan, A Hayat Khan, S Haider, R Mahmood Journal of renewable and Sustainable Energy 4 (3)	163	2012
Shifting of agro-climatic zones, their drought vulnerability, and precipitation and temperature trends in Pakistan S Adnan, K Ullah, S Gao, AH Khosa, Z Wang International Journal of Climatology 37, 529-543	140	2017

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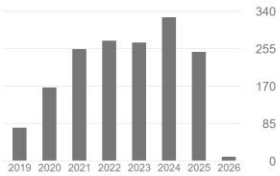

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Pakistan Council for Science & Technology
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S.No	Name/ Designation/ Department/ Organization	Sci ID	Books Published/ Edited	Patents	Ph.Ds Supervised/ Co-Supervised	External Research Grants	Total Publications	Impact Factor	Citations	Citations/ Article	*h*-index	Awards	Innovations/ Applied Research Output	Grand Score
87	**Mrs. Almas Hamid** Assistant Professor Environmental Sciences & Chemistry, Kinnaird College for Women, Lahore	**28442**	0	0	0	1	5	7.223	65	13.000	2	0	0	**1.785**
88	**Prof. Dr. Nasir Ahmad** Professor Institute of Geology, University of the Punjab, Lahore	**496**	0	1	16	0	0	0	0	0	0	0	0	**1.776**
89	**Dr. Muhammad Basharat** Associate Professor Institute of Geology, University of Azad Jammu & Kashmir, Muzaffarabad	**28297**	0	0	0	0	11	12.749	39	3.545	4	0	0	**1.773**
90	**Prof. Dr. Muhammad Hassan Agheem** Professor Centre For Pure & Applied Geology, University of Sindh, Jamshoro	**29200**	0	0	0	1	5	14.053	41	8.200	3	0	0	**1.725**
91	**Dr. Shaheen Begum** Assistant Professor Environmental Sciences, Fatima Jinnah Women University, Rawalpindi	**2666**	0	0	0	0	8	14.276	59	7.375	3	0	0	**1.709**
92	**Dr. Audil Rashid** Assistant Professor Department of Environmental Sciences, PMAS Arid Agriculture University, Rawalpindi	**2179**	0	0	2	0	10	15.625	28	2.800	3	0	0	**1.697**
93	**Dr. Hussani Mubarak** Assistant Professor Department of Soil and Environmental Sciences, Faculty of Agricultural Sciences, Ghazi University, Dera Ghazi Khan	**29936**	0	0	0	0	5	10.004	23	4.600	4	0	0	**1.647**
94	**Dr. Aamir Ali** Assistant Professor Department of Earth Sciences, Quaid-i-Azam University, Islamabad	**2945**	0	0	0	1	13	10.468	30	2.308	3	0	0	**1.598**
95	**Dr. Muhammad Ishaq Kakar** Associate Proofessor Centre of Excellence in Mineralogy, University of Balochistan, Quetta	**3174**	0	0	0	1	9	15.77	27	3.000	3	0	0	**1.584**
96	**Dr. Khuram Shahzad Ahmad** Assistant Professor Department of Environmental Sciences, Fatima Jinnah Women University, Rawalpindi	**28786**	0	0	0	1	9	7.057	37	4.111	3	0	0	**1.557**
97	**Dr. Arshad Ashraf** Principal Scientific officer Climate change, Alternate energy and Water Resource Institute (CAEWRI), Pakistan Agricultural Research Council, Islamabad	**28034**	0	0	0	0	5	6.949	29	5.800	3	0	0	**1.552**
98	**Mr. Shahzada Adnan** Meteorologist National Drought Monitoring Centre,Pakistan Meteorological Department, COMSATS Institute of Information Technology, Islamabad	**28797**	0	0	0	0	5	11.248	23	4.600	3	0	0	**1.503**
99	**Dr. Sohaib Aslam** Assistant Professor Environmental Sciences, Forman Christian College University, Lahore	**28015**	0	0	0	0	4	11.598	24	6.000	3	0	0	**1.490**
100	**Mr. Amer Jamal Hashmat** SENIOR SCIENTIST Environmental Biotechnology Division, NIBGE, Pakistan Atomic Energy Commission, Faisalabad	**29816**	0	0	0	0	2	4.694	24	12.000	2	0	0	**1.474**
101	**Dr. Khaista Rehman** Assistant Professor National Centre of Excellence in Geology, University of Peshawar, Peshawar	**30217**	0	0	0	1	9	8.462	20	2.222	3	0	0	**1.432**
102	**Dr. Saman Shahid** Assistant Professor Sciences and Humanities, National University of Computer & Emerging Sciences, Lahore	**29539**	0	0	0	0	9	21.711	15	1.667	2	0	0	**1.398**
103	**Dr. Muhammad Iqbal** Manager Scientific Research Space Applications & Research Complex, Pakistan Space and Upper Atmosphere Research Commission, Islamabad	**28037**	0	0	0	1	2	6.259	13	6.500	2	0	0	**1.379**

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For the Paper

Investigations into Precipitation and Drought Climatologies in South Central Asia with Special Focus on Pakistan over the Period 1951-2010

Published in

Journal of Climate 2016,29,6019-6035

Authored by

Shahzada Adnan, Kalim Ullah, Gao Shouting

March 01, 2018
Date

Prof. Dr. Mukhtar Ahmed
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Higher Education Commission

Dr. MUHAMMAD AFZAAL
DIRECTOR
Pakistan Meteorological Department
Islamabad

COMSATS University Islamabad

Ref.No.MET/CUI/721

May 6th, 2018

Dear Shahzada Adnan
PhD Scholar,
Department of Meteorology,
COMSATS University Islamabad,
Islamabad.

Subject: **Research Productivity Awards (2015-2017)**

I am writing to express my sincere congratulations on being awarded the Research Productivity Awards from 2015 to 2017 on publishing the following peer review international research articles.

- **Adnan, S.,** Ullah, K., & Gao, S. (2015). Characterization of drought and its assessment over Sindh, Pakistan during 1951–2010. *Journal of Meteorological Research*, *29*(5), 837-857. **(IF: 1.116)**
- **Adnan, S.,** Ullah, K., & Shouting, G. (2016). Investigations into precipitation and drought climatologies in South Central Asia with special focus on Pakistan over the period 1951–2010. *Journal of Climate*, *29*(16), 6019-6035. **(IF: 4.161)**
- **Adnan, S.,** Ullah, K., Gao, S., Khosa, A. H., & Wang, Z. (2017). Shifting of agro-climatic zones, their drought vulnerability, and precipitation and temperature trends in Pakistan. *International Journal of Climatology*, *37*, 529-543. **(IF: 3.10)**

This prestigious award recognizes your outstanding research productivity and significant contributions to the field of Meteorology. This award is a well-deserved recognition of your hard work, intellectual curiosity, and commitment to pushing the boundaries of knowledge. It exemplifies the exceptional caliber of research being conducted by our PhD students.

As Head of Department, I am delighted with your achievements. Being a good researcher you are a valuable asset to our research community. Wish you all the best for your future research endeavor.

Sincerely,

Dr. Kalim Ullah
Head
Tenured Associate Professor
Department of Meteorology
COMSATS University Islamabad

Dr. MUHAMMAD AFZAAL
DIRECTOR
Pakistan Meteorological Department
Islamabad